TELIPORT
Script Your Future Season 4

TATA
TATA ELXSI

CASE STUDY 1 / CONNECTED FLEET PREDICTIVE MAINTENANCE

FlairGuard AI

Maintenance decisions when telemetry is missing, unreliable or constrained by service capacity

Team: _____ Members: _____ College: _____ Team ID: _____

CAUSAL OBSERVATIONS TO AUDITABLE MAINTENANCE ACTIONS

Telemetry + features
21 sensors, 3 settings
Past-only rolling windows

→

RUL + anomaly models
LightGBM / HGB / Ridge
Regime-aware preprocessing

→

Telemetry trust
Missingness, staleness
Persistence and abstention

→

Rule-based triage
RUL and quality thresholds
Human review for exceptions

→

Orders + execution
Priority, deduplication
SQLite audit and recovery

MATLAB research worker: similarity-based RUL, matched Python comparisons and versioned evidence

TRUST

Fit-only normalization and engine-disjoint evaluation. Causal histories prevent future-data leakage. Missing telemetry remains explicit.

VALUE

FD002 trigger outage: regime imputation produced 0/80 modeled failures versus 14/80 with bounded countdown. Scenario-specific result.

EXECUTION

FastAPI + React operator workflow. Persisted work orders and model pins. Genuine retraining, rollback and backend restart verified.

332 full-life policy runs independently scored across 169 source engines. MATLAB comparison runs through the application.

Retrospective NASA C-MAPSS study with hypothetical maintenance costs. Advanced-policy superiority and fresh model promotion remain unestablished.